Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
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Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My goal is to make ionic compounds and to show the shape of that chemical compound.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

https://app.molview.com/ *The website above generates a 3D movable diagram when you design a certain compound.*

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

The purpose of this website is to help students learn about ionic compound and learn about how they bond. It will help put chemical compounds into perspective and model them in real life.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The ultimate goal of this program is to take in a chemical element using user input, process its most popular bonding patterns by processing data, and output the Lewis Structure and shape of the compound.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will input a metal they wish to form a compound with from a specific list provided to the user.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The computer will evaluate that data by verifying that the element is valid, producing the Lewis Structure of that specific element, taking user input using a button to bond the element, if the compound is not stable, having another option to add more elements, producing the Lewis structure of the compound, then taking user input if they want to see the overall shape of the compound.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get a model of what the compound looks like with a Lewis Structure and the option to output a 3D shape of the molecule.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things"

(nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

The core things that my program needs to track is “Metal”, “Non-metal”, “Lewis Structure”, and “3D Model.”

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

For “Metal” specific details that I need are it’s name, symbol, ion charge, and number of electrons. Details that I will abstract away is its atomic weight and number of protons.

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

I think the most difficult part of the project will be outputting images on drafter. We have never had to generate images on drafter, so it will be a challenge.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will most likey being using an LLM to help code especially for the imaging aspect. I will ensure that I understand the code by running through it and documenting it. I will not use code from the LLM that I do not understand or use code that is too advanced based on our knowledge of coding skills.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My goal for this project is to better understand drafter from working start to finish.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I asked my friend, "I want to make a tool to visualize the creation of compounds. What helps you understand how compounds form?"

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

My friend suggested modeling valence electrons for each compound. She said that modeling the lewis structures would be the most helpful while modeling this program.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

The most ambitious feature is modeling any possible ionic compound. This is too complicated for an inductury programming project because the range of compounds is too vast and the bonding of compound is too unpredictable when it comes to exceptions of elements. My friend suggested to just model a few compounds. She said to model the most popular compounds therefore the modeling of the compounds is more predictable.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

Based on this conversation, my project has become more focused to only model the most popular compounds.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If i get stuck, I could refer back to previous projects, and if I am still stuck, I could ask a friend for guidance. If they too don't understand, them I will try to attend office hours for guidance. My last resort would be using an LLM for guidance.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

My step-by-step debugging plan is to trace the program line by line and try to find the error. If I cannot point out the error, I will ask the LLM to trace the code and see if they can point out the error.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

N/A. I did not use an LLM for this assignment.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.